

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS: Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

1. Responsive Web Design (RWD)

Responsive Web Design (RWD) is a web development approach that enables a website to automatically adjust its layout and content according to the screen size of the device being used. A responsive website provides an optimal viewing experience on desktops, tablets, and

smartphones without requiring separate versions of the site.

Importance of Responsive Web Design

- Provides a consistent user experience across devices.
- Eliminates horizontal scrolling and layout issues.
- Improves website accessibility and readability.
- Enhances SEO, as search engines favor mobile-friendly websites.
- Reduces development and maintenance costs by using a single responsive design.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

a) Viewport Meta Tag

The viewport meta tag allows the webpage to adapt to the width of the user's device.

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

b) Flexible Units

Instead of fixed pixel values, use flexible units such as %, em, rem, vw, and vh.

```
.container{  
  
    width:100%;  
  
}
```

c) Responsive Images

Images should resize automatically without overflowing.

```
img{  
  
    max-width:100%;  
  
    height:auto;  
  
}
```

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Flexbox is a one-dimensional layout model used to arrange items in rows or columns. It is ideal for creating responsive navigation bars, content sections, and sidebars

```
<div class="container">
```

```
<div class="content">
```

Main Content

```
</div>
```

```
<div class="sidebar">
```

Sidebar

```
</div>
```

```
</div>
```

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

1. Use a mobile-first design approach.
2. Optimize images to reduce loading time.
3. Use CSS Grid for complex page layouts.
4. Add smooth transitions and animations.
5. Ensure buttons and links are large enough for touch devices.
6. Improve accessibility using semantic HTML and ARIA labels.
7. Test the website on different browsers and devices.

Code of Conduct:

I Ch. Bhargav Ram (192425334) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: